INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

MODULE-1

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PROJECT NAME: HotelRevAI AI Driven Revenue Analysis for Hotels_Batch -13

About the Project:

HotelRev Analyze is a data analytics project developed to evaluate and monitor hotel performance using booking and revenue data. The project focuses on analyzing important hotel performance metrics such as Occupancy Rate, Average Daily Rate (ADR), and Revenue Per Available Room (RevPAR) to understand how efficiently hotel rooms are being utilized and how much revenue they generate.

The project involves preparing booking data in Microsoft Excel, transforming it using Power Query, and creating calculated metrics and dashboards in Power BI. Through this process, raw booking data is converted into meaningful insights that help identify trends in hotel demand, pricing strategies, and seasonal performance.

The final output of the project is an interactive Power BI dashboard that visually represents hotel revenue performance through charts, KPI cards, and tables. These insights help hotel management understand occupancy trends, compare revenue across time periods, and make data-driven decisions to improve profitability.

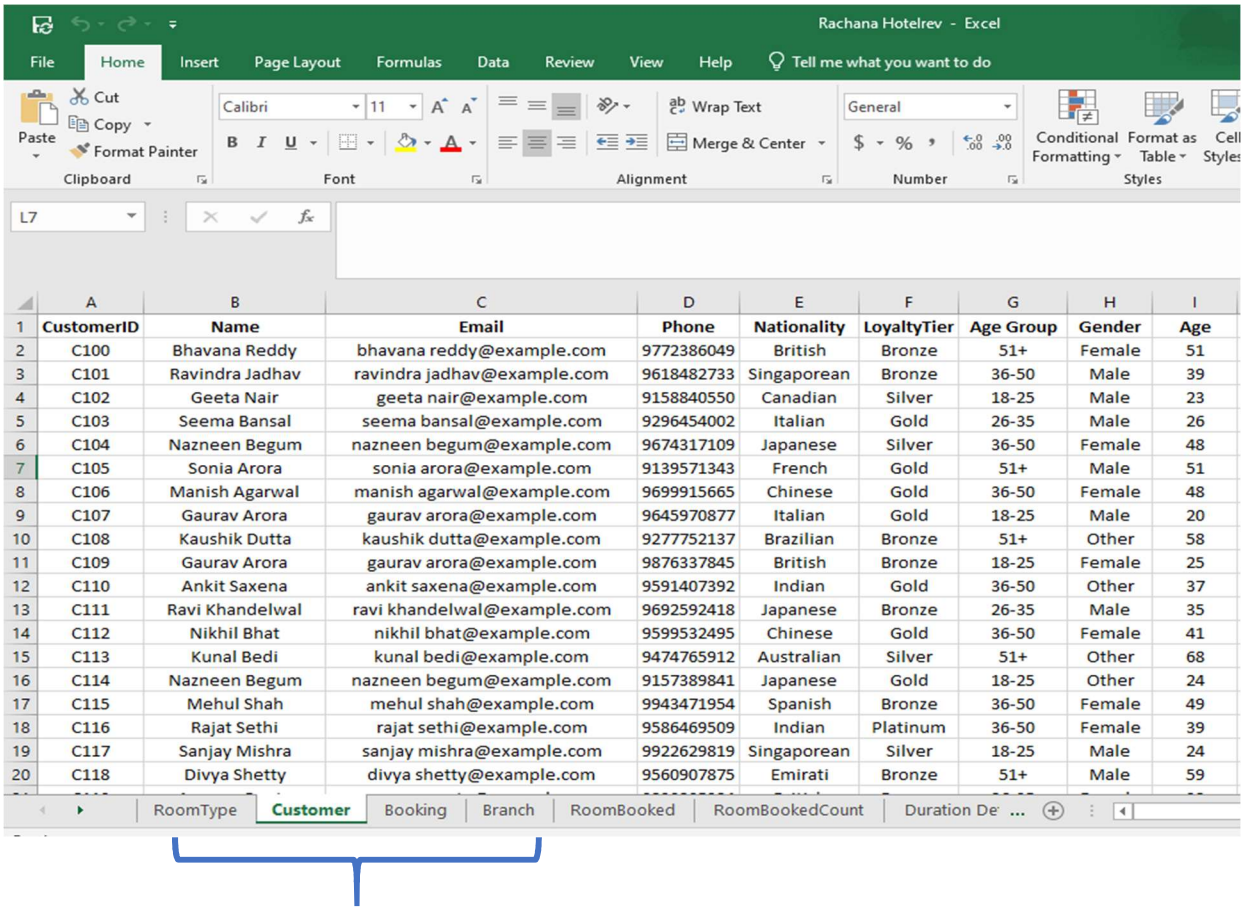
MODULE 1:INTRODUCTION AND DATA PREPARATION

Objective:-

The objective of Module 1 is to generate a realistic hotel booking dataset using Python. Synthetic data is created for customers, hotel branches, room types, and bookings to simulate real hotel operations. The generated data is exported into CSV files. These datasets serve as the foundation for further data transformation and analysis in Power BI.

Step-1: GENERATE BASIC DATASETS

The datasets were generated using a Python script that simulates hotel booking operations. Random data was created for customers, hotel branches, room types, and booking transactions using Python libraries such as **csv,** random, and datetime. The generated data was then exported as CSV files to be used for further analysis in Power BI.



CustomerID	Name	Email	Phone	Nationality	LoyaltyTier	Age Group	Gender	Age
C100	Bhavana Reddy	bhavana reddy@example.com	9772386049	British	Bronze	51+	Female	51
C101	Ravindra Jadhav	ravindra jadhav@example.com	9618482733	Singaporean	Bronze	36-50	Male	39
C102	Geeta Nair	geeta nair@example.com	9158840550	Canadian	Silver	18-25	Male	23
C103	Seema Bansal	seema bansal@example.com	9296454002	Italian	Gold	26-35	Male	26
C104	Nazneen Begum	nazneen begum@example.com	9674317109	Japanese	Silver	36-50	Female	48
C105	Sonia Arora	sonia arora@example.com	9139571343	French	Gold	51+	Male	51
C106	Manish Agarwal	manish agarwal@example.com	9699915665	Chinese	Gold	36-50	Female	48
C107	Gaurav Arora	gaurav arora@example.com	9645970877	Italian	Gold	18-25	Male	20
C108	Kaushik Dutta	kaushik dutta@example.com	9277752137	Brazilian	Bronze	51+	Other	58
C109	Gaurav Arora	gaurav arora@example.com	9876337845	British	Bronze	18-25	Female	25
C110	Ankit Saxena	ankit saxena@example.com	9591407392	Indian	Gold	36-50	Other	37
C111	Ravi Khandelwal	ravi khandelwal@example.com	9692592418	Japanese	Bronze	26-35	Male	35
C112	Nikhil Bhat	nikhil bhat@example.com	9599532495	Chinese	Gold	36-50	Female	41
C113	Kunal Bedi	kunal bedi@example.com	9474765912	Australian	Silver	51+	Other	68
C114	Nazneen Begum	nazneen begum@example.com	9157389841	Japanese	Gold	18-25	Other	24
C115	Mehul Shah	mehul shah@example.com	9943471954	Spanish	Bronze	36-50	Female	49
C116	Rajat Sethi	rajat sethi@example.com	9586469509	Indian	Platinum	36-50	Female	39
C117	Sanjay Mishra	sanjay mishra@example.com	9922629819	Singaporean	Silver	18-25	Male	24
C118	Divya Shetty	divya shetty@example.com	9560907875	Emirati	Bronze	51+	Male	59

Customer.csv:

- CustomerID
- Name
- Email
- Phone
- Nationality
- Loyality tier
- Gender
- Age

Booking.csv:

- BookingID
- CustomerID
- RoomTypeID
- BranchID
- CheckInDate
- CheckOutDate
- Duration

- Revenue
- BookingStatus
- CancellationReason
- LeadTime
- PaymentMethod
- DiscountApplied
- BookingChannel

Branch.csv:

- BranchID
- BranchName
- City
- Phone
- ManagerName
- StarRating
- HasPool
- HasConferenceRoom

RoomType.csv:

- RoomTypeID
- RoomType
- Price
- MaxOccupancy
- Amenities
- IsAC
- IsSeaFacing

Step-2: Creating Additional coulmns required for Analysis.

In Booking.csv we have added staytype, RoomNightSold So that we can use it for accurate analysis.

	M	N	O	P	Q	R
1	PaymentMethod	DiscountApplied	BookingChannel	Purpose	Stay Type	RoomNightSold
2	UPI	10	Travel Agent	Business	Short Stay	2
3	UPI	10	Call Center	Vacation	Day Use	0
4	Cash	20	Website	Vacation	Short Stay	2
5	Credit Card	10	Call Center	Vacation	Extended Stay	0
6	Corporate Account	20	Call Center	Business	Short Stay	0
7	UPI	0	Website	Holiday	Extended Stay	0

StayType Formula:

=IF(H2=1,"Day Use",

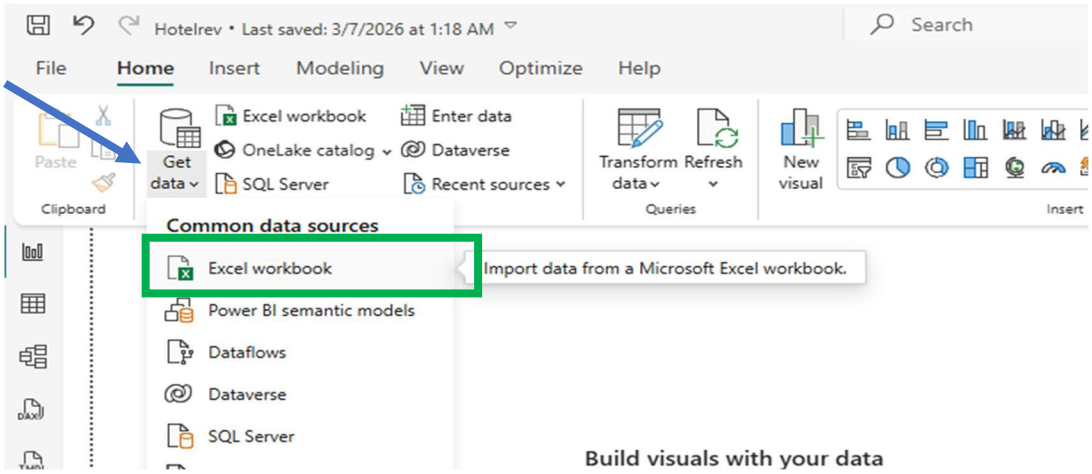
```
IF(H2<=2,"Short Stay",
  IF(H2<=5,"Extended Stay",
    "Long Stay")))
```

RoomNightSold Formula:

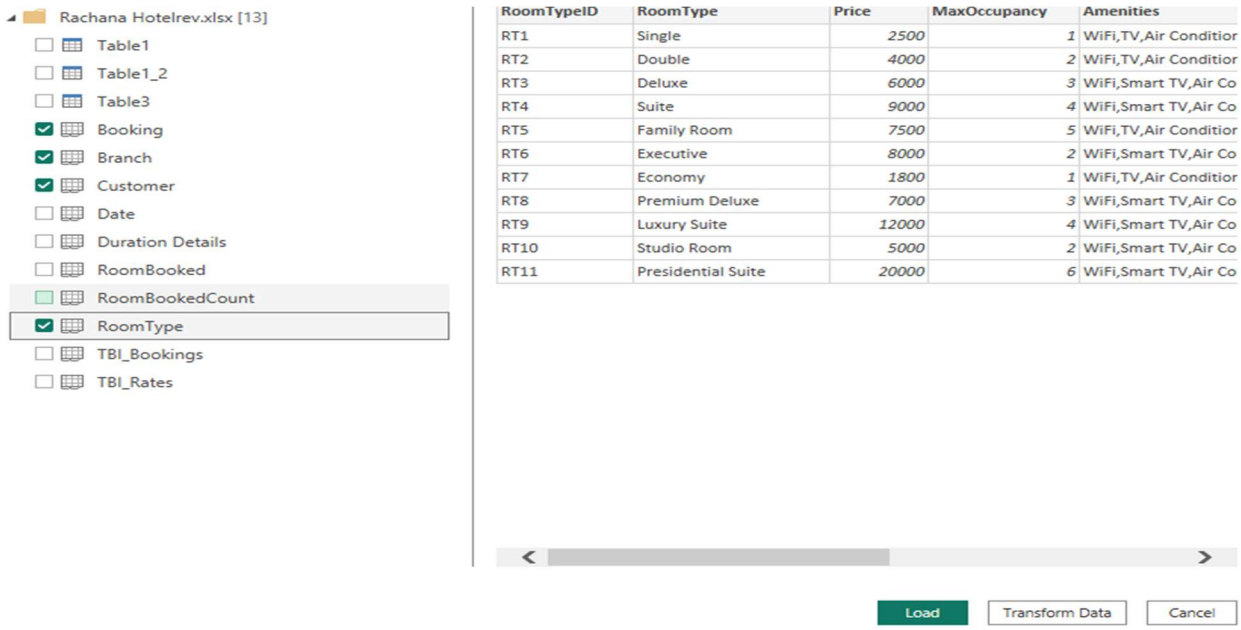
=SUMIFS(H2, J2, "Checked-In")

Step-3: Importing the data into PowerBI

The prepared dataset files were imported into Power BI using the Get Data option. This step allows the data to be loaded into Power BI for further transformation, modeling, and visualization.



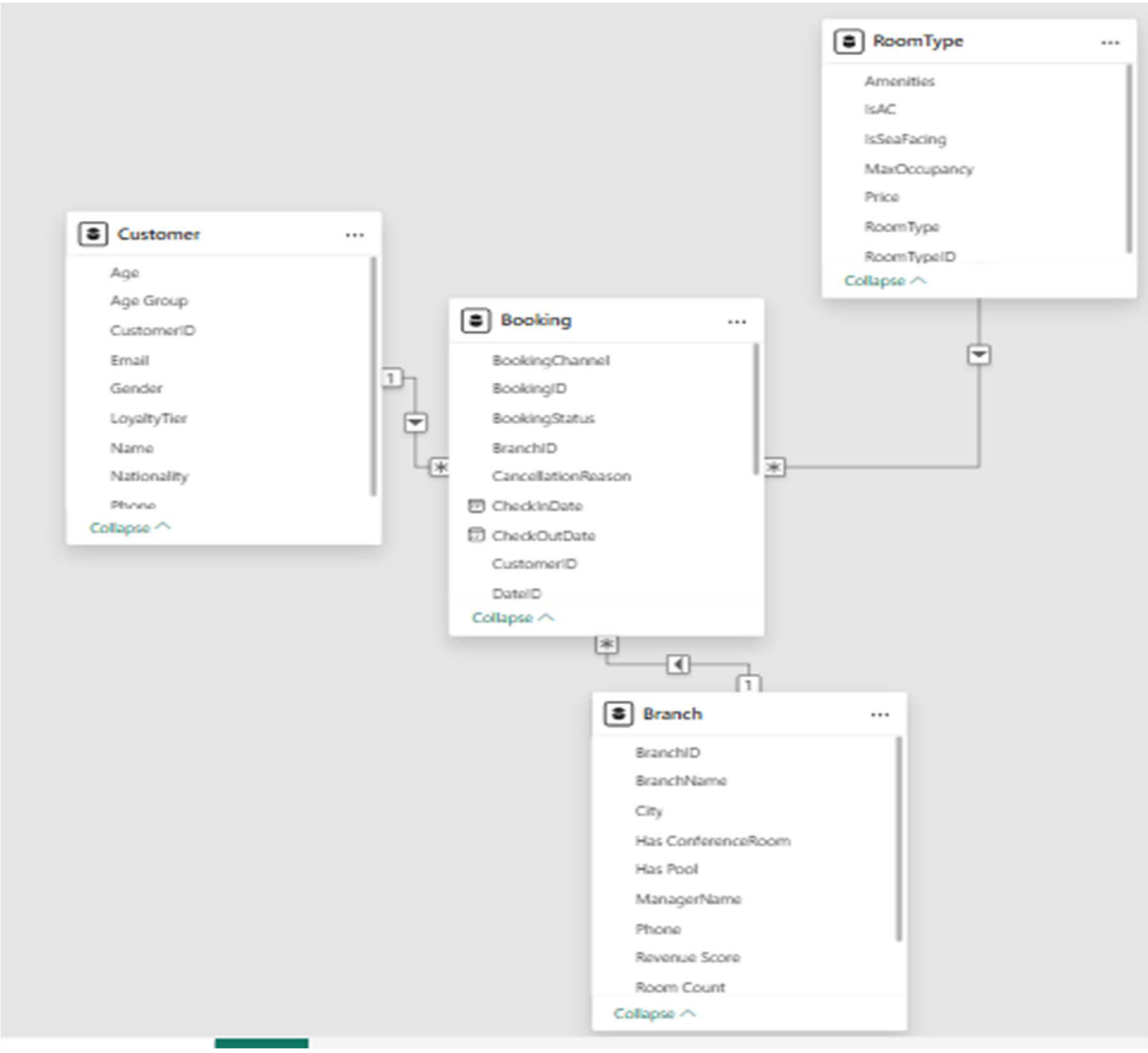
Select necessary tables and then click load



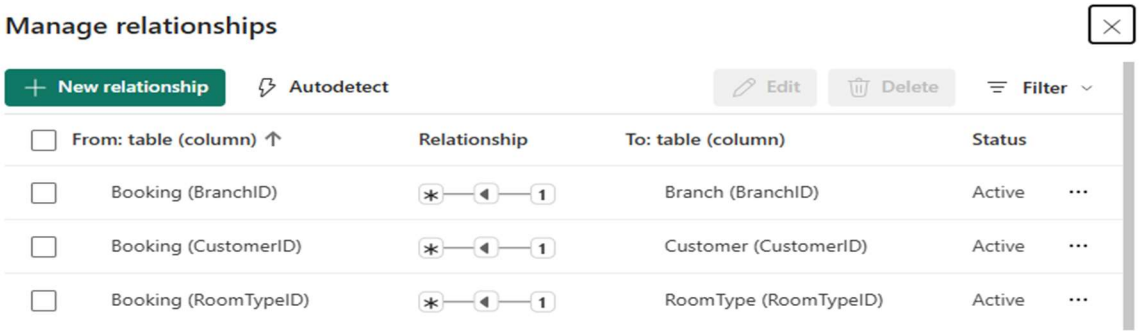
RoomTypeID	RoomType	Price	MaxOccupancy	Amenities
RT1	Single	2500	1	WiFi,TV,Air Conditior
RT2	Double	4000	2	WiFi,TV,Air Conditior
RT3	Deluxe	6000	3	WiFi,Smart TV,Air Co
RT4	Suite	9000	4	WiFi,Smart TV,Air Co
RT5	Family Room	7500	5	WiFi,TV,Air Conditior
RT6	Executive	8000	2	WiFi,Smart TV,Air Co
RT7	Economy	1800	1	WiFi,TV,Air Conditior
RT8	Premium Deluxe	7000	3	WiFi,Smart TV,Air Co
RT9	Luxury Suite	12000	4	WiFi,Smart TV,Air Co
RT10	Studio Room	5000	2	WiFi,Smart TV,Air Co
RT11	Presidential Suite	20000	6	WiFi,Smart TV,Air Co

Step-3: Create STAR schema in PowerBI

After importing the datasets, a Star Schema data model was created in Power BI by connecting the fact table (Booking) with related dimension tables such as Customer, Branch, and RoomType. This structure helps organize the data efficiently and improves performance for analysis and reporting.



Relationships between the tables as given in the following figure:



Conclusion:

In this module, the required datasets for the hotel revenue analysis project were successfully generated using Python. Additional columns needed for analysis were created using Excel formulas, and the prepared datasets were imported into Power BI. Finally, a star schema data model was created to organize the data efficiently for further transformation, analysis, and visualization.